



# TRUSTWORTHY, PRIVACY-PRESERVING, AND FUNCTIONAL DATA OUTSOURCING SYSTEMS

**Ph. D. Candidate:** Tung Le

**Advisor:** Dr. Thang Hoang

Computer Science Department,

Virginia Tech

March 20, 2026

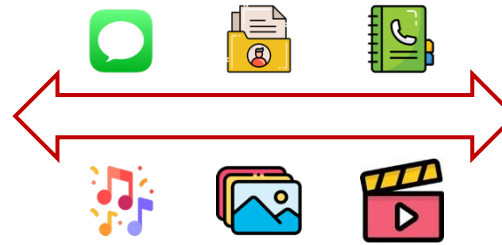
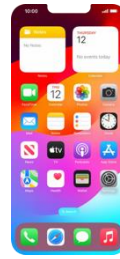
**Ph. D. Committee:**

1. Dr. Thang Hoang (Chair)
2. Dr. Lenwood S. Heath
3. Dr. Daphne Yao
4. Dr. Wenjing Lou
5. Dr. Rouzbeh Behnia



# Overview: Cloud Storage

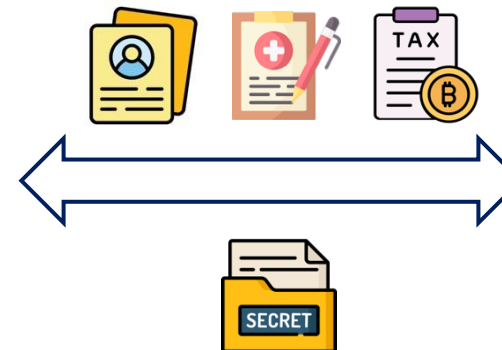
Handheld device



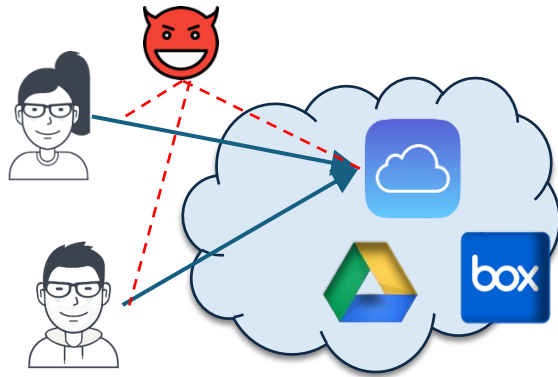
Personal device



Government  
Corporate  
Hospital



## Cloud storage:



**Data misuse!**

**Data breach!**

**Swedish healthcare advice line stored 2.7 million patient phone calls on unprotected web server**

The Swedish healthcare advice line stored 2.7 million patient phone calls on unprotected web server.

By Tammy

4746  
Brea  
\$5732  
\$696  
06  
\$52A  
808B4  
00F2A  
20736  
719\$  
028BE

**Payroll data for 29,000 Facebook employees stolen**

Banking information, Social Security digits, names and salaries were stolen from several unencrypted hard drives left in a payroll employee's car, a report says.

Corinne Reichert  
Dec. 13, 2019 2:21 p.m. PT



The payroll data on 29,000 Facebook was reportedly stolen off unencrypted hard drives.

**Fortune 500 company leaked 264GB in client, payment data**

Updated: The data leak impacted Tech Data's client servers, SAP systems, and more.

**Russian Government Hackers Penetrated DNC, Stole Opposition Research on Trump**

By Ellen Nakashima, Washington Post



## Encrypted cloud storage:



An advertisement for encrypted cloud storage. It features logos for Icedrive, PREVEIL, Tresorit, and pCloud. Text includes 'The next generation', 'Secure and simple to use cloud storage for your documents', and 'Backcloudation, easy'.

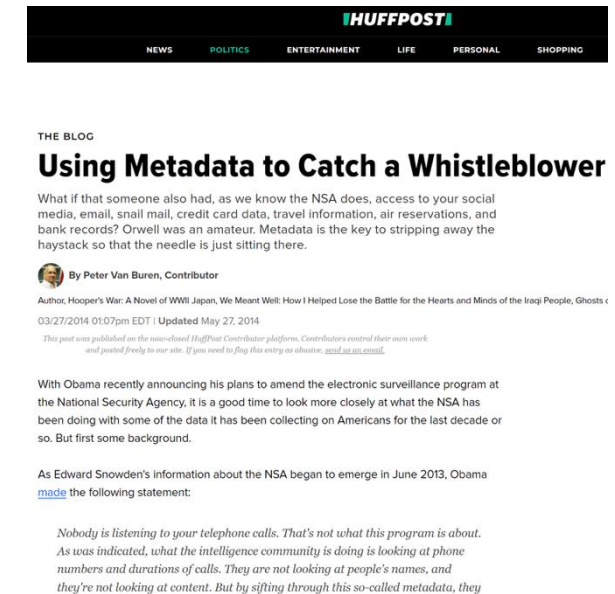
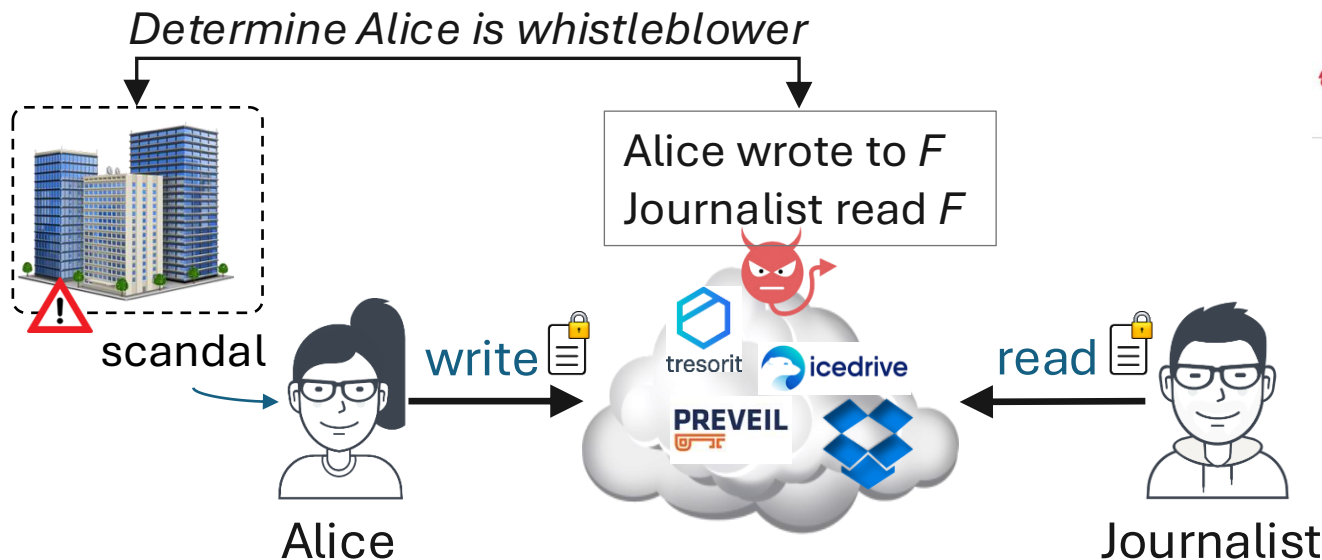
# Is Data Confidentiality Enough?

Data is always kept encrypted, however:

- Data integrity and soundness are still concerns

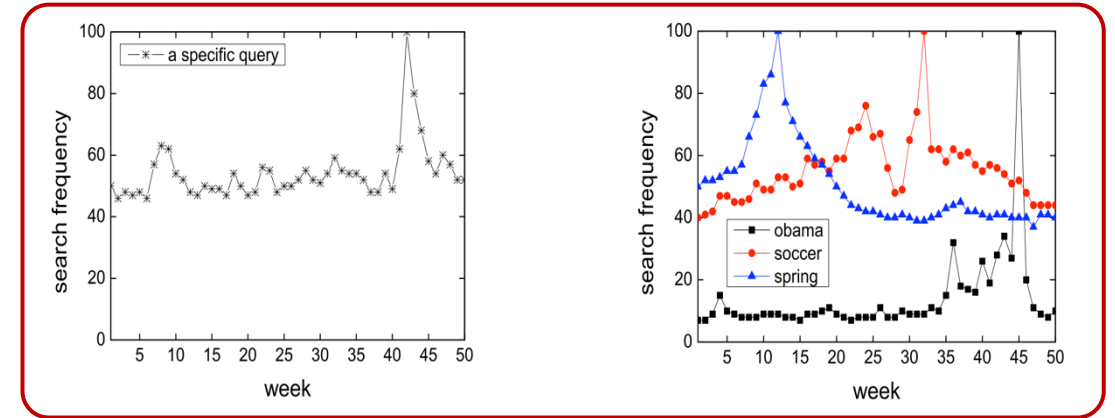
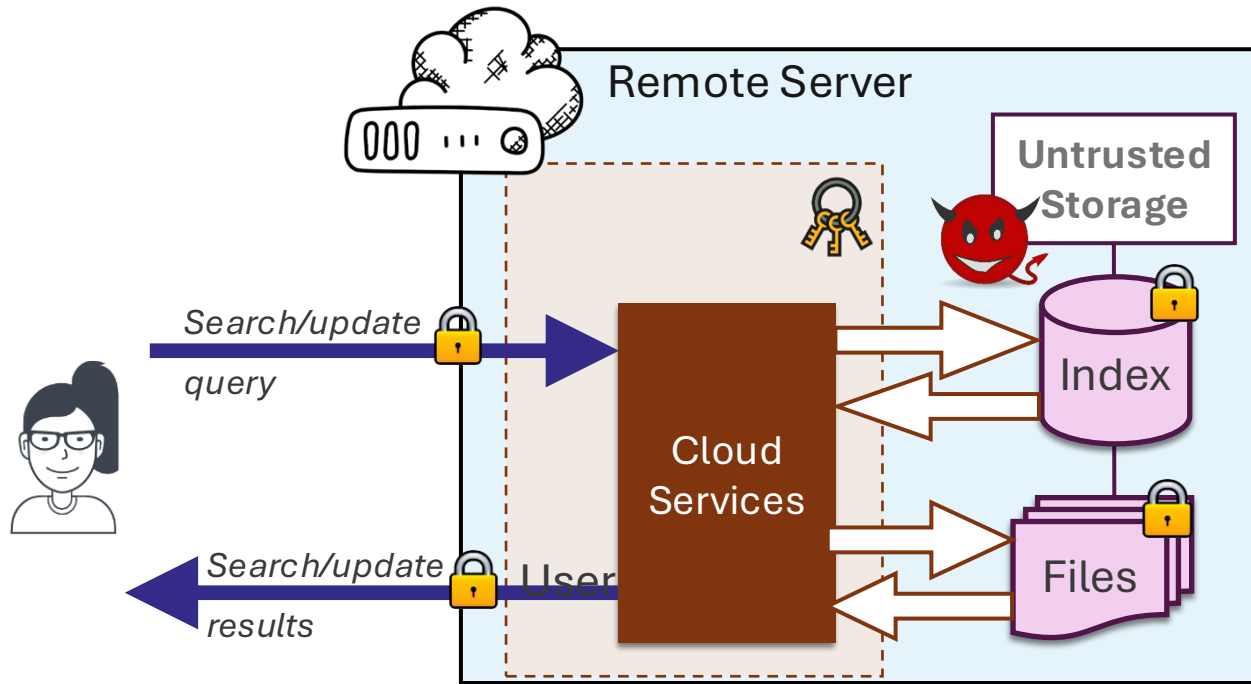


- Sensitive information can still be inferred from **metadata**

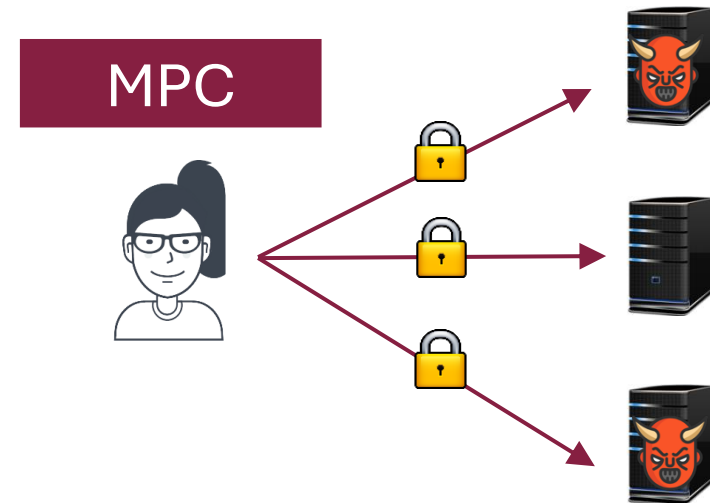


# Functionalities on Encrypted Data

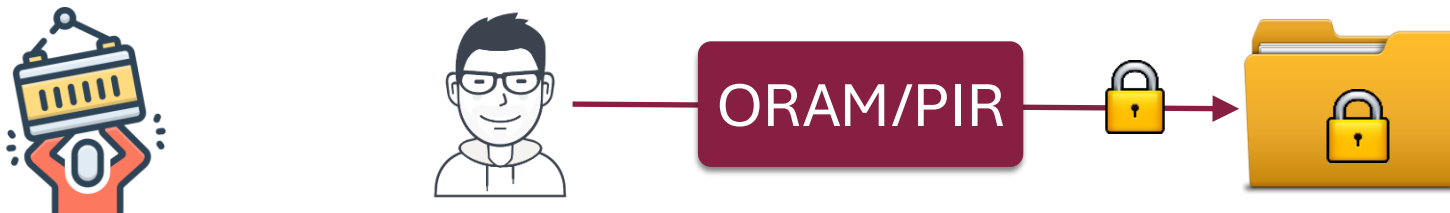
- How to efficiently support secure functionalities on encrypted data?



## Statistical inference analysis

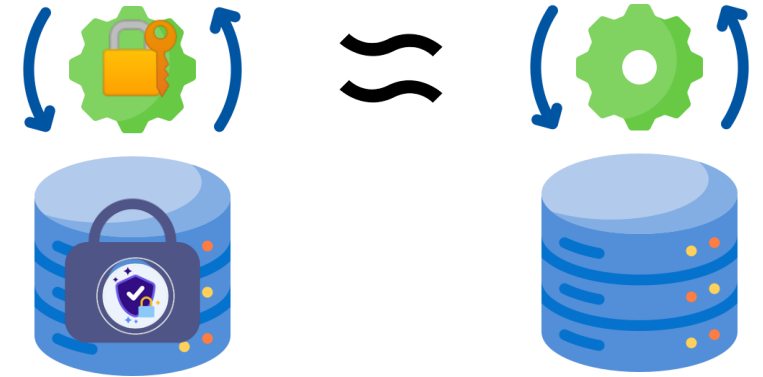


## Costly cryptographic primitives:



Trustworthy data outsourcing services are expected to: 

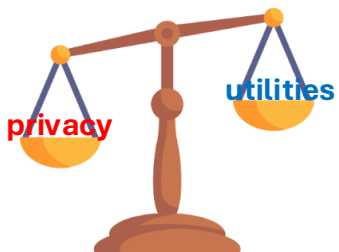
1. **Keep** user **data intact**



2. **Ensure** data and user **privacy**



3. **Provide** essential **functionalities**: keyword search



The goal of my dissertation is to **efficiently** resolve **security**, **privacy**, and **functionalities** issues **simultaneously** in data outsourcing systems



## Chapter I: Data Intactness

- Propose an **efficient technique** for **verifying** the **integrity** of **data**



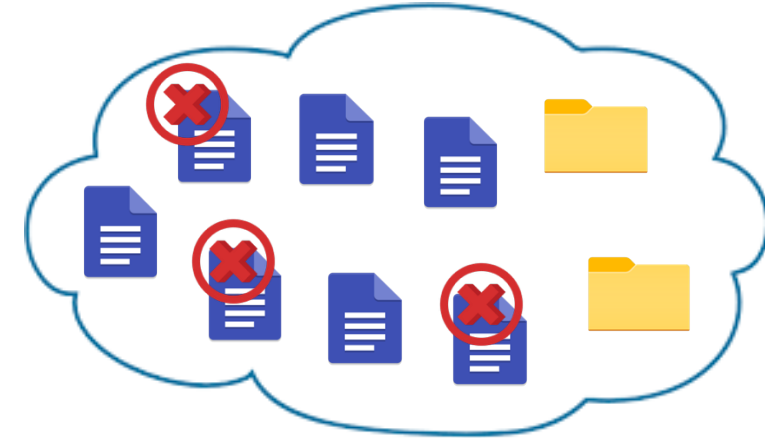
## Chapter II: Privacy-Preserving Data Actions

- Focus on **keyword search** functionality with **security** and **efficiency**
- Propose a series of **searchable encryption techniques** achieving high **security** guarantees and **performance** in various threat and system models





## Authenticated Storage

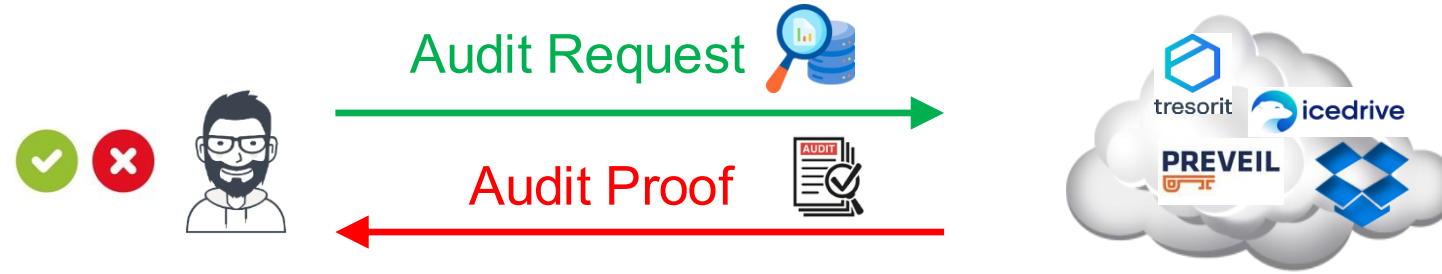


**Data loss** can happen due to:

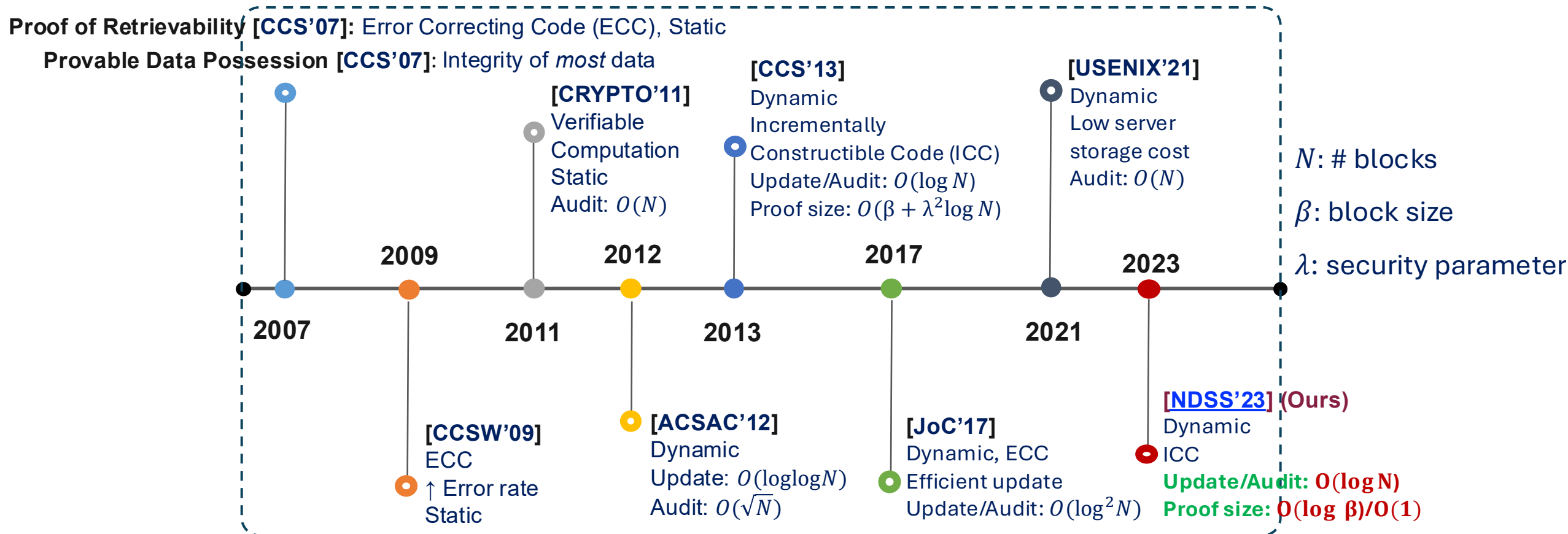
- Hardware failures
- Adversarial behaviors



# Two Decades of Proof of Retrievability (PoR)



First proposal



Despite two decades of development, auditing remains a costly operation!

- **Research Gap:** 

- Minimal audit cost: audit proof size and end-to-end audit latency

- **Our Porla [NDSS'23]:** 

- *Minimize* audit cost:

$N$ : #data blocks

✓ Audit bandwidth:  $O(\log \beta)$  or  $O(1)$  , where:  $\beta$ : data block size

✓ Server/Client:  $O(\lambda \log N)$

$\lambda$ : security parameter

- Maintain a reasonable update performance:

✓ Server:  $O(\log N)$

✓ Client/Bandwidth:  $O(\beta)$



## Main Techniques:



- Incrementally Constructible Code
- Our Homomorphic MAC

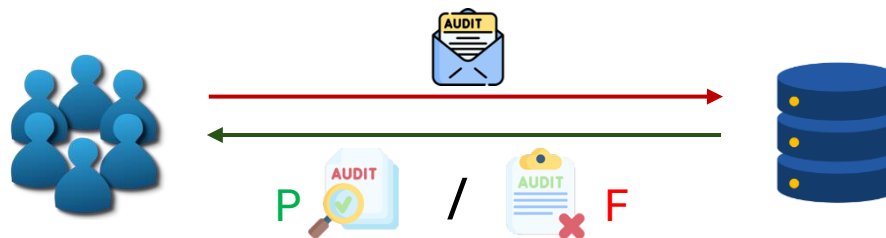


$$c_1 \times \text{yellow tag} + c_2 \times \text{red tag} + c_3 \times \text{blue tag} = \text{grey tag}$$

- Verifiable Computation Techniques



- Support Public Audit



- $87 \times -14,012 \times$  smaller proof size than previous DPoR schemes

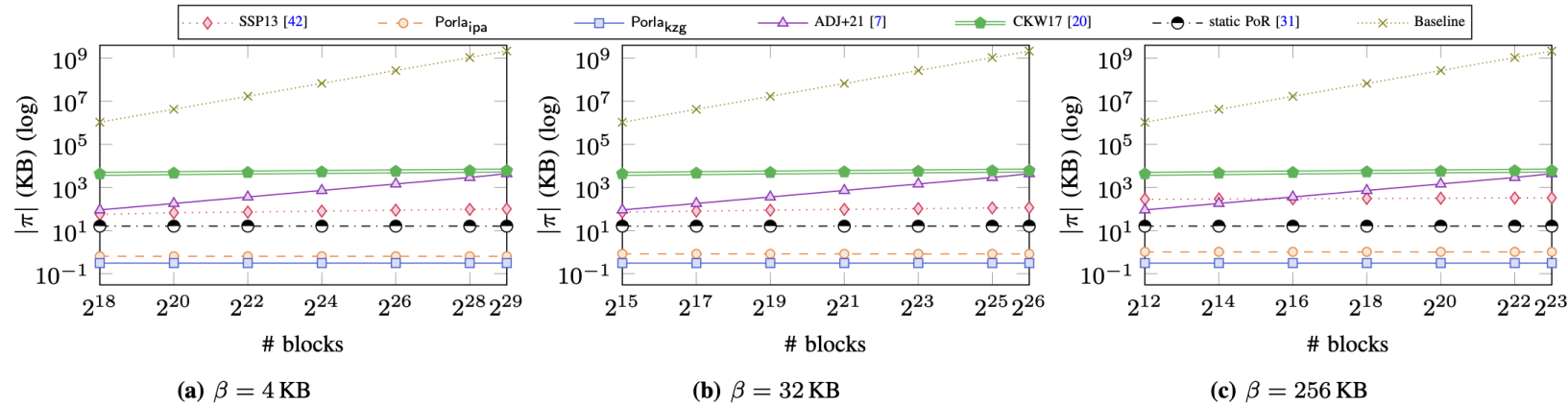


Fig. 6: Audit proof size of Porla and its counterparts.

- $4 \times -18,000 \times$  faster audit time than prior approaches

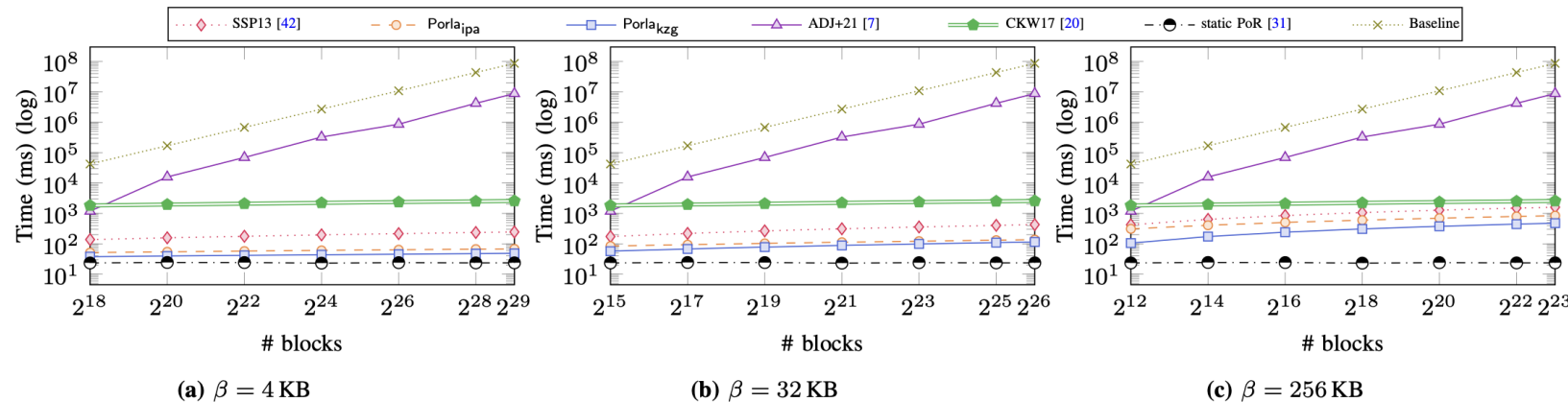


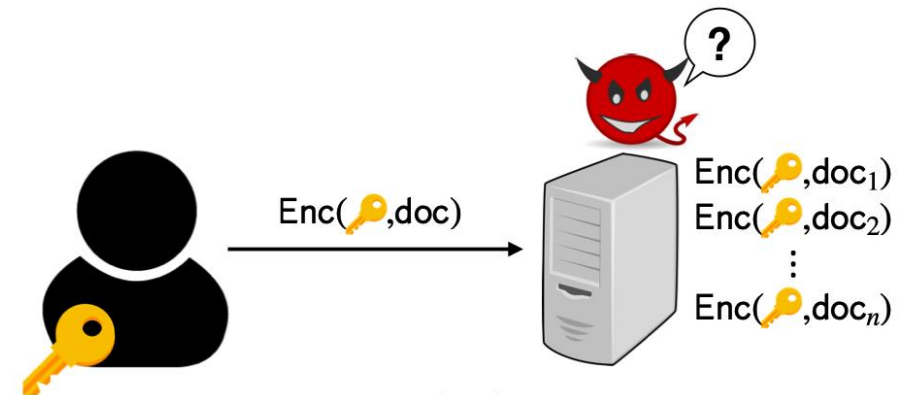
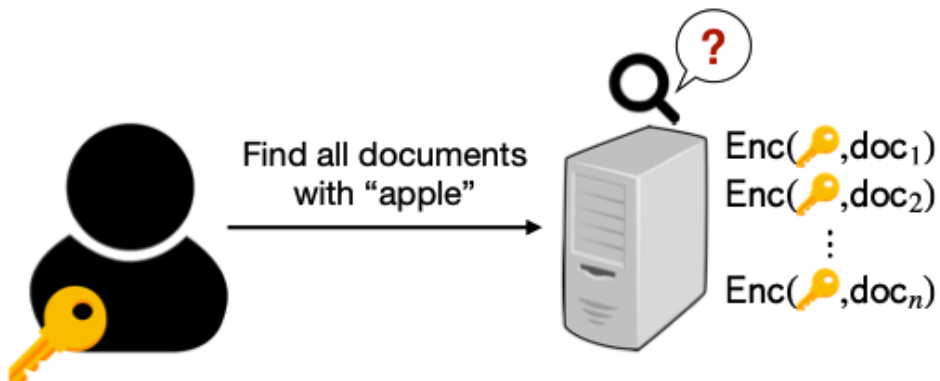
Fig. 7: End-to-end audit delay of Porla and its counterparts.

Users expect the ability to execute search

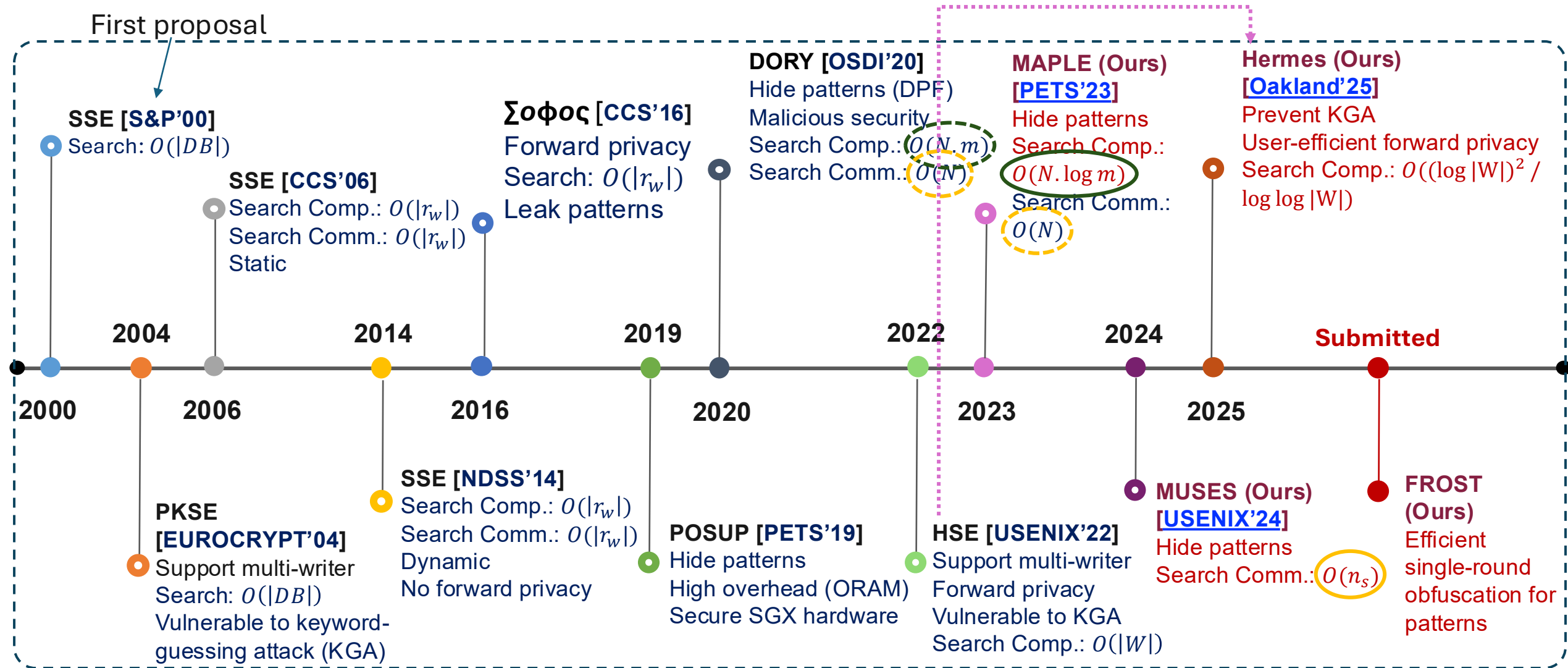
Apple

Doc 1  
Doc 7  
Doc 21  
Doc 53

**Challenge:** server cannot decrypt data to search



# 20 Years+ of Searchable Encryption (SE)

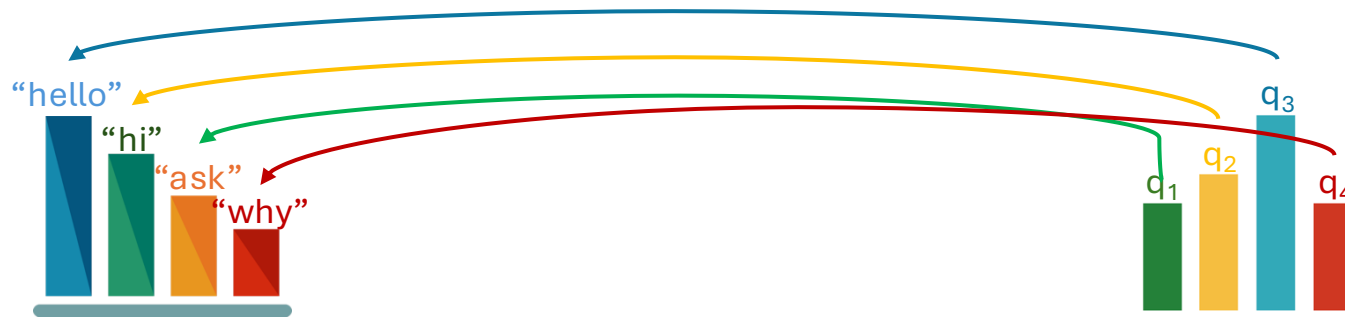


Improve communication, computation efficiency, and security

- **Frequency Attack [IS'14]:**

Auxiliary information

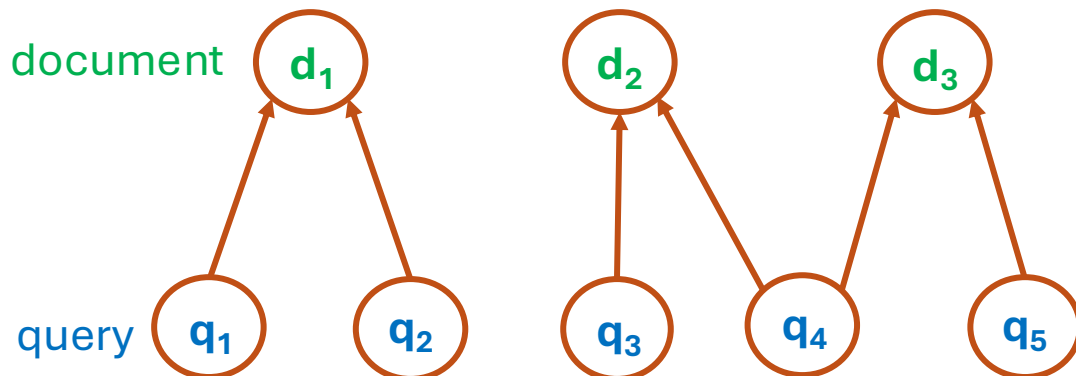
Query-leakage information



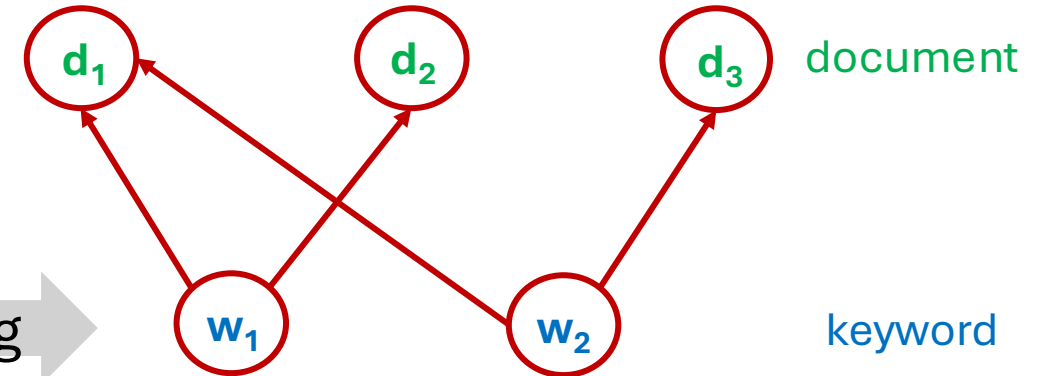
Attack can be formalized as a graph matching problem. For example:

- **Subgraph<sup>VL</sup>/Subgraph<sup>ID</sup> [NDSS'20]:** bipartite graph matching

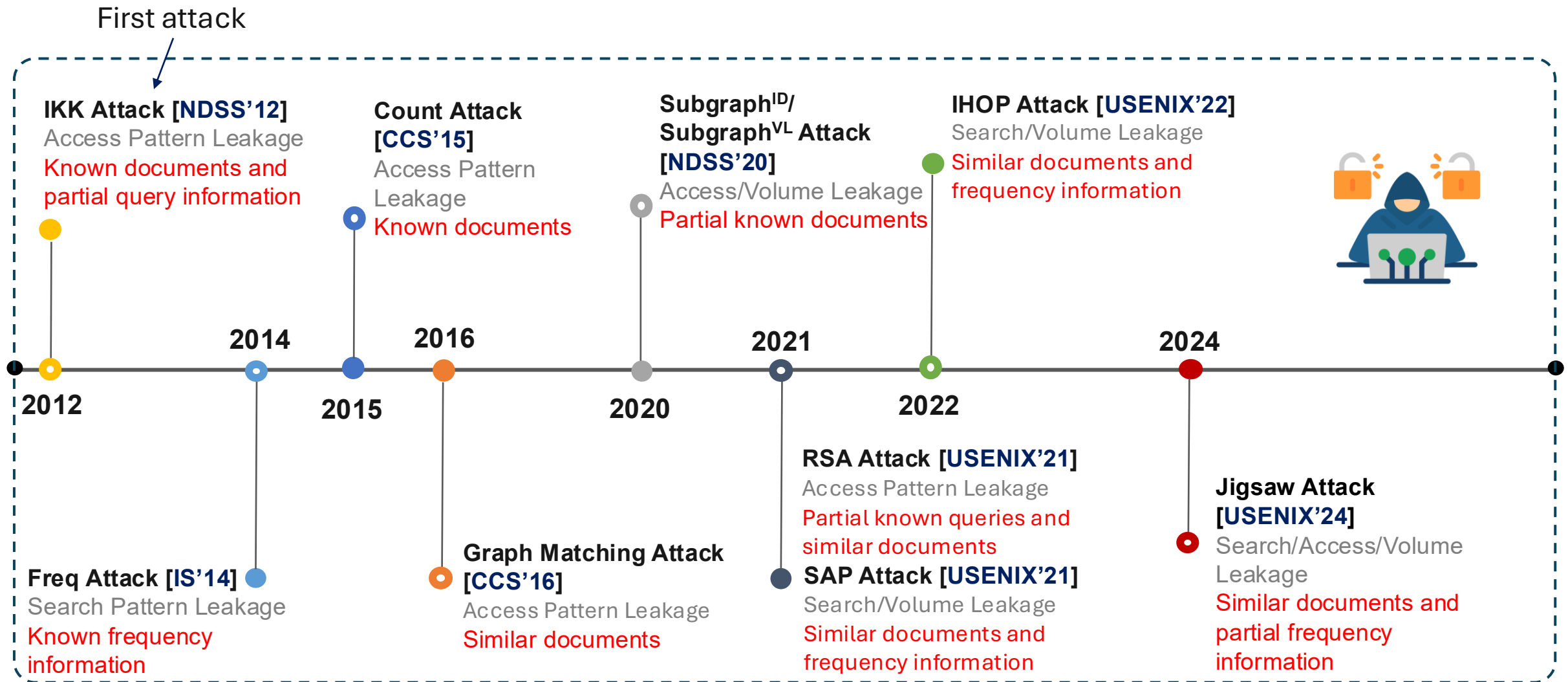
Graph of observed query tokens (tags)



Graph of keywords in auxiliary data



Matching



*Attacks become more effective under more realistic assumptions*

## Research Gap:



- Search complexity with pattern-hiding is  $O(N \cdot m)$ , where:

$N$ : #documents, and  $m$ : keyword representation size

- Limited multi-user support: assume all users are trusted or control access policies based on access level



## Our **MAPLE** [PETS'23]:

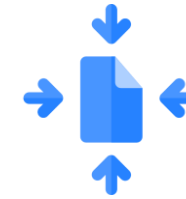
- Server search complexity:  $O(N \cdot \log m)$
- **Hide all metadata**: search, result and volume patterns
- Multi-user with **fine-grained access control**



## Main techniques:



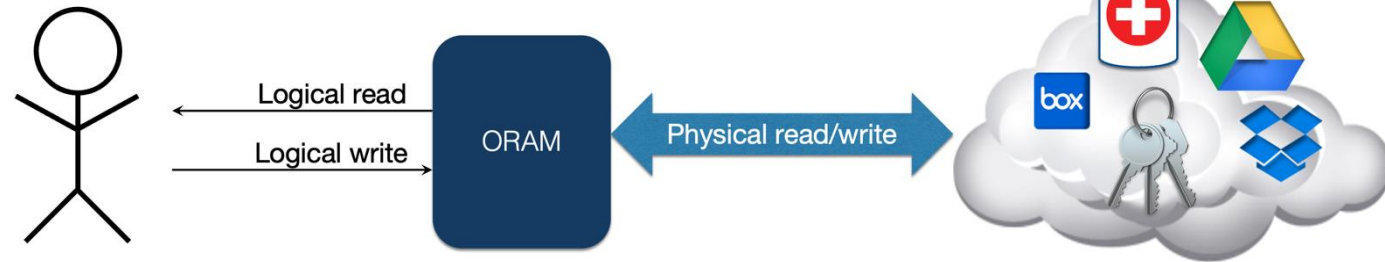
- Bloom Filter to compress search index
- Oblivious Random Access Memory (ORAM)



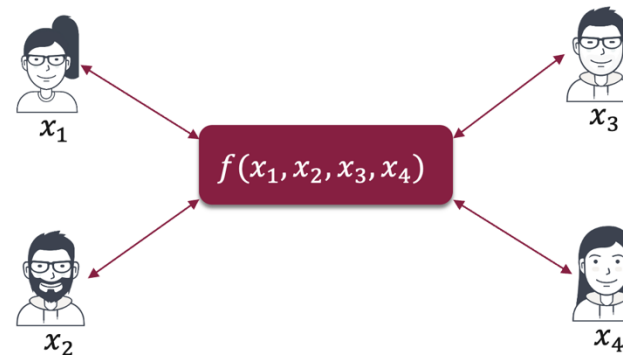
|       | a<br>"amazon" | G<br>"google" | N<br>"netflix" | ... | Apple<br>"apple" |
|-------|---------------|---------------|----------------|-----|------------------|
| doc 1 | 1             | 0             | 1              | 0   | 1                |
| doc 2 | 0             | 1             | 1              | 0   | 0                |
| doc 3 | 1             | 1             | 0              | 0   | 0                |
| ...   | 0             | 1             | 0              | 1   | 1                |
| doc N | 1             | 0             | 0              | 1   | 1                |

Bitmap for keywords in doc 2

- Circuit ORAM
- Oblivious Table



- Multiparty Computation



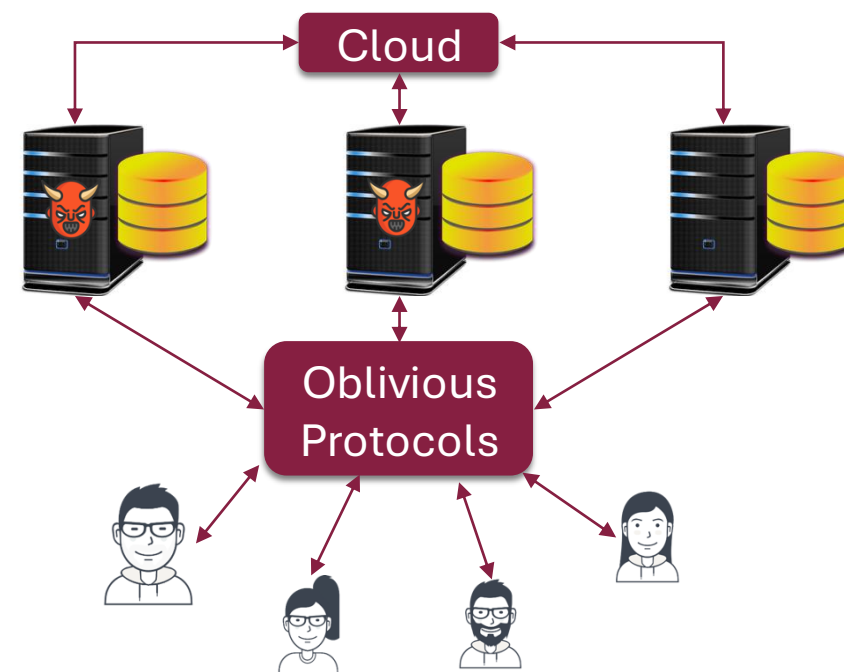
- Generic MPCs are powerful but expensive



- Distributed computations specifically designed for a particular computation task are more efficient

Our **MUSES** [USENIX'24]:

- **Hide all statistical information:**  
search, result, and volume patterns
- **Minimal user overhead** for search and permission revocation



## Main techniques:

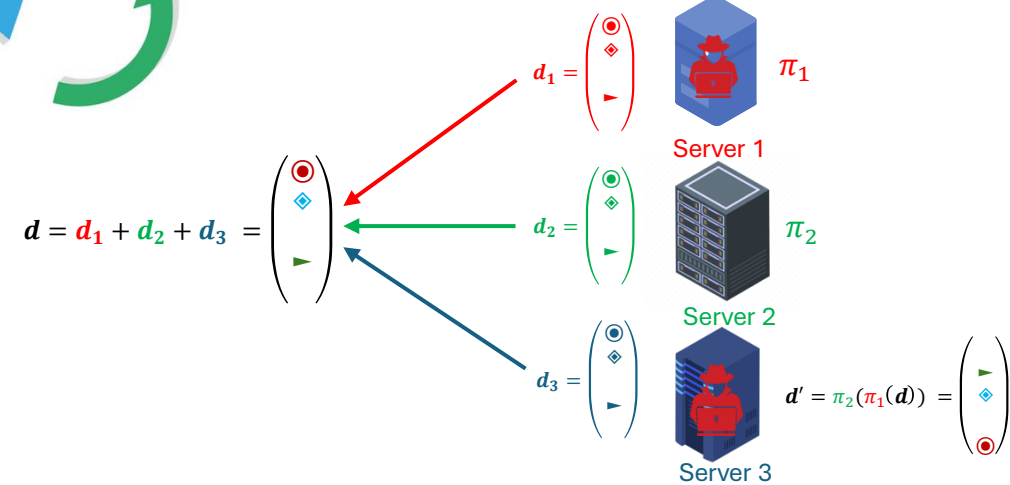
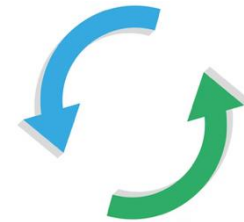
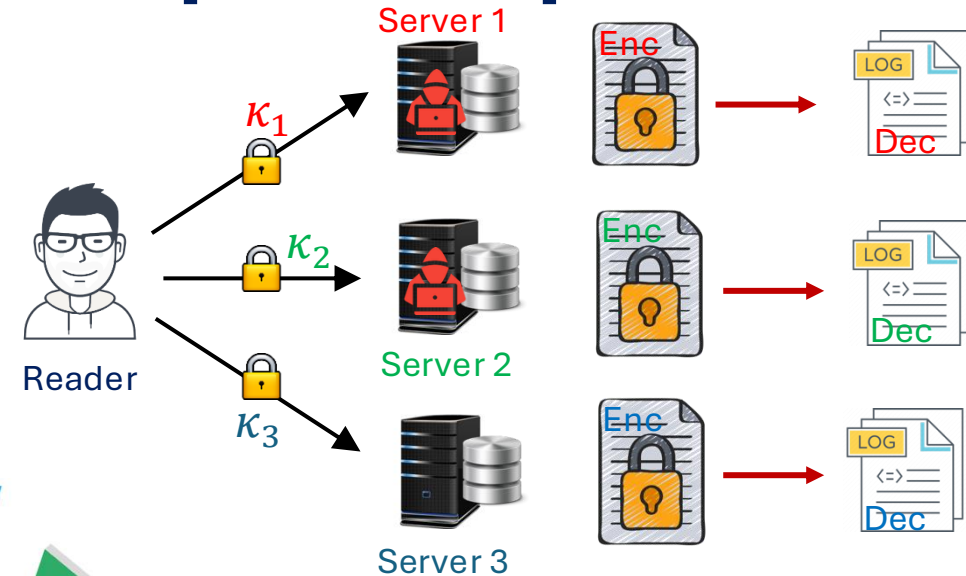
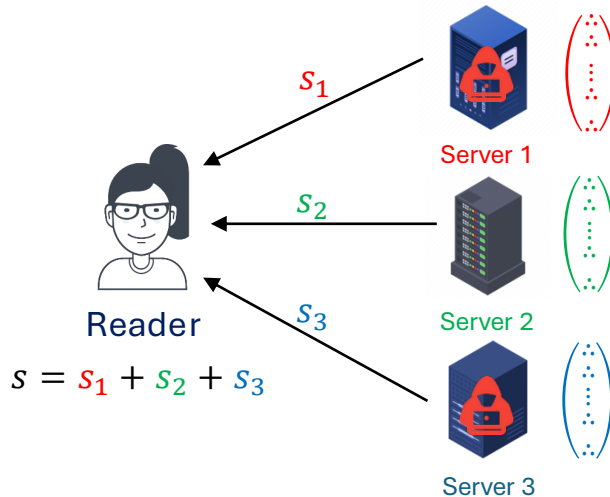


- Key-Homomorphic Pseudorandom Function [CRYPTO'13]



Delegating Decryption

- Our Multiparty Oblivious Counting



- Our Multiparty Oblivious Shuffling

- Security of Multiparty Oblivious Counting ( $\Pi_{\text{Loc}}$ ):

**Theorem 1.**  $\Pi_{\text{Loc}}$  protocol is perfectly semi-honest secure in the TSS-hybrid model.

- Security of Multiparty Oblivious Shuffling ( $\Pi_{\text{Los}}$ ):

**Theorem 2.** Assuming that the adversary can statically corrupt at most  $L - 1$  out of the  $L$  parties,  $\Pi_{\text{Los}}$  is  $\mathcal{L}$ -secure in the TSS-hybrid model, where  $\mathcal{L}_{\Pi_{\text{Los}}.\text{Prep}} = \{\emptyset\}$ , and  $\mathcal{L}_{\Pi_{\text{Los}}.\text{Shffl}} = \{\pi(\mathbf{d}) \wedge \mathcal{P}_L \in I_C\}$ , where  $\pi(\mathbf{d})$  is a pseudorandom permutation of input vector  $\mathbf{d}$ , and  $I_C$  is the set of  $L - 1$  corrupted parties.

## Security Proof: Hybrid Model

Real World



Executing the real protocol



Ideal World



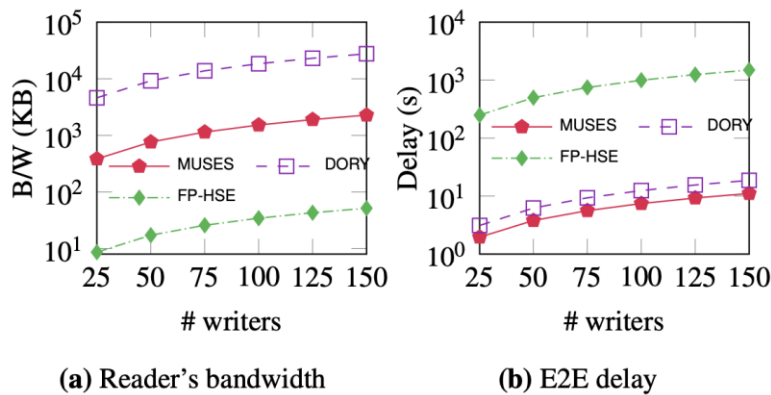
Simulating with random data

## Reader's bandwidth:

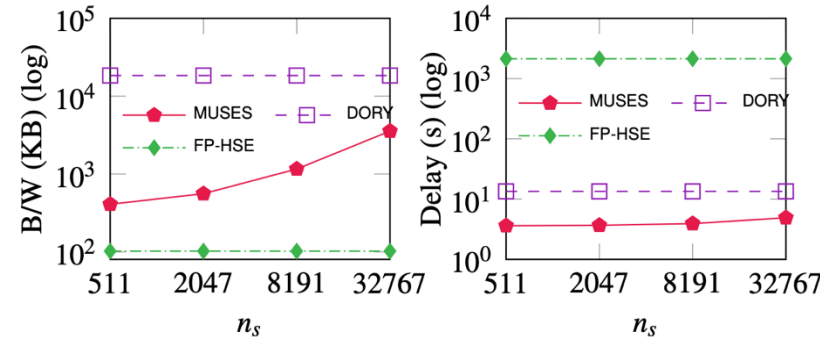
**$12 \times - 97 \times$**  smaller than DORY (hide patterns),  **$6 \times$**  larger than FP-HSE (leak patterns)

## End-to-end latency:

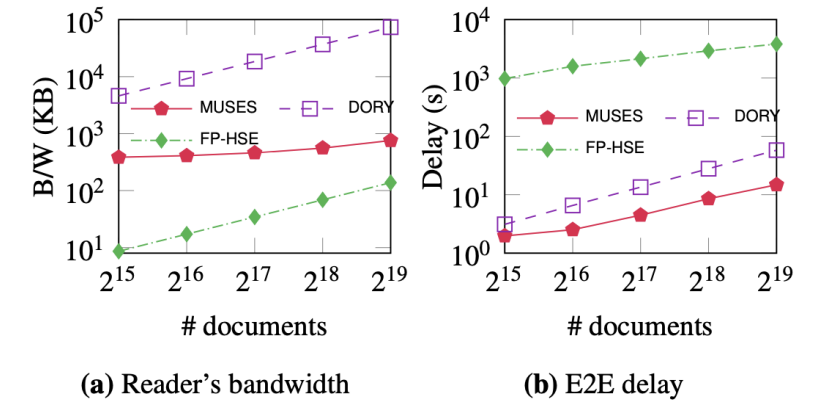
**$2 \times - 4 \times$**  faster than DORY,  **$127 \times - 632 \times$**  faster than FP-HSE



**Figure 6:** Keyword search performance (log scale on y-axis).



**Figure 10:** Keyword search performance with varying  $n_s$ .



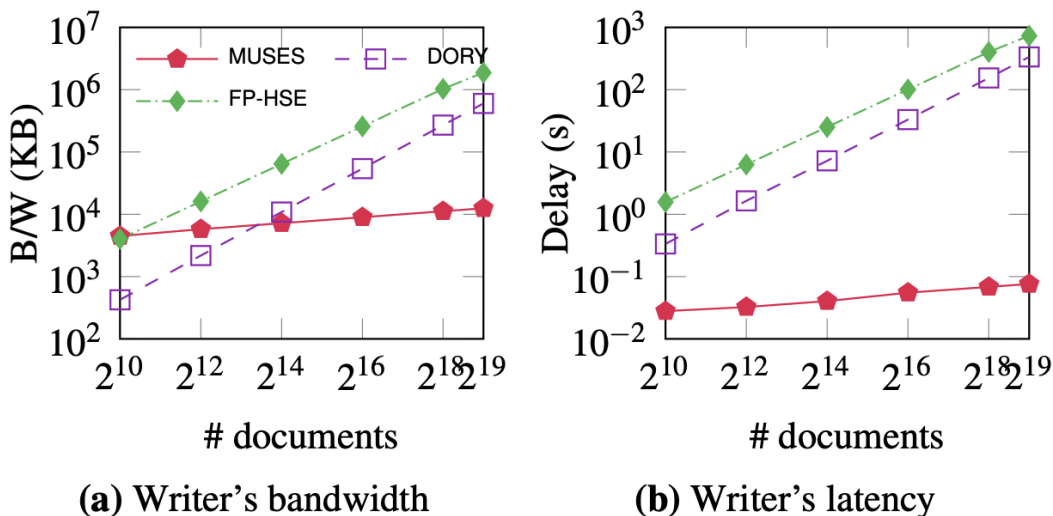
**Figure 11:** Keyword search performance w/ varying database sizes.

## Writer's bandwidth:

**$2 \times -150 \times$**  smaller than DORY/FP-HSE

## End-to-end latency:

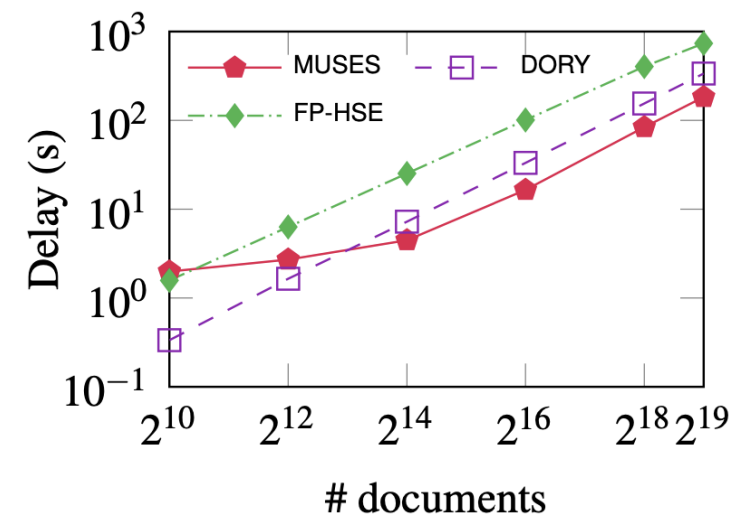
**$2 \times -6 \times$**  faster than DORY/FP-HSE



**Figure 7:** Permission revocation performance (log scale on y-axis).

## Writer's latency:

**$12 \times -9600 \times$**  faster than DORY/FP-HSE



**Figure 8:** E2E permission revocation delay (log scale on y-axis).

- MUSES and MAPLE rely on distributed computation for secure search
  - Expensive deployment and maintenance cost
- PKSE [EUROCRYPT'04] and HSE [USENIX'22] can support multi-user more naturally in practical settings (e.g., email and messaging)
- Several open problems:
  - Keyword-guessing attacks
  - Inefficient forward privacy
  - High server computation cost for search



This work addresses the above fundamental security and performance issues

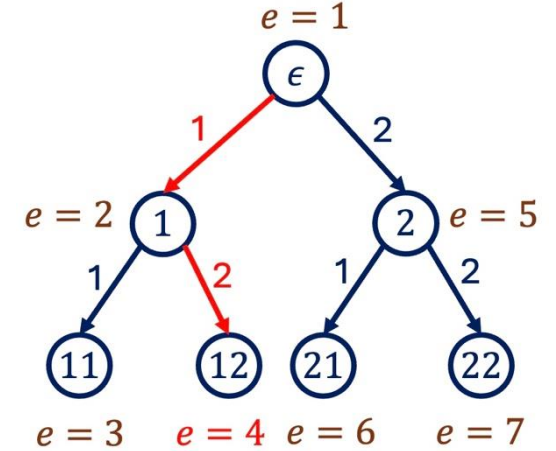
- **Idea 1:** Hidden ID-Coupling Key Aggregate Encryption

- Prevent keyword-guessing attacks



- **Idea 2:** Tree-Based Epoch Encoding

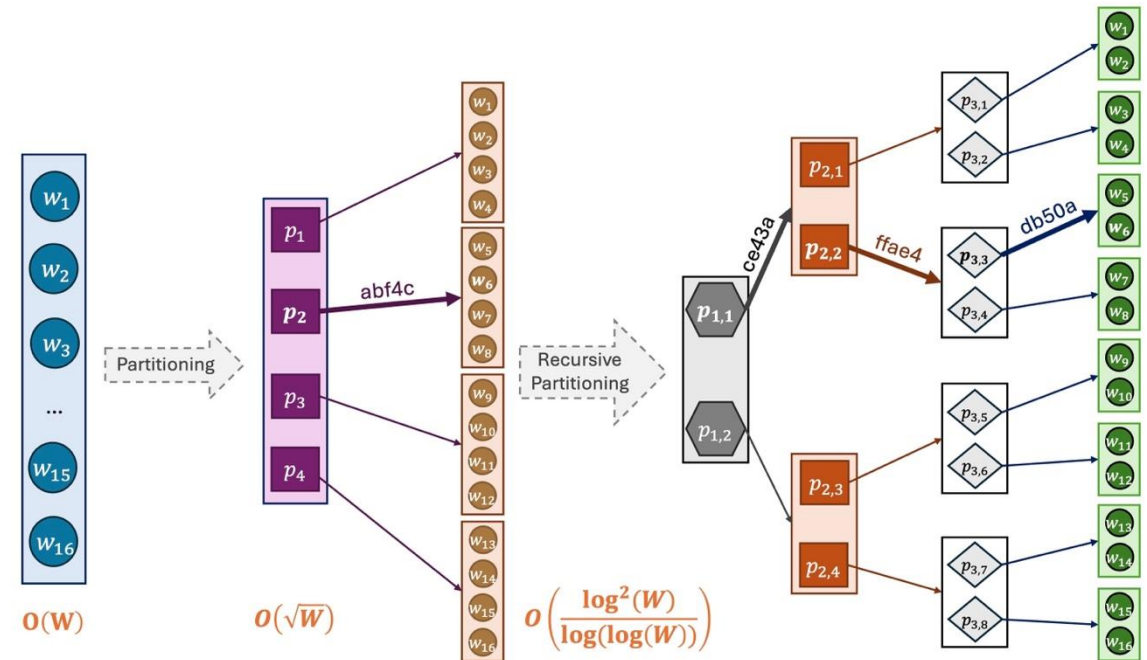
- Achieve epoch-based **forward privacy** with **user efficiency**



- **Idea 3:** (Recursive) Partitioning

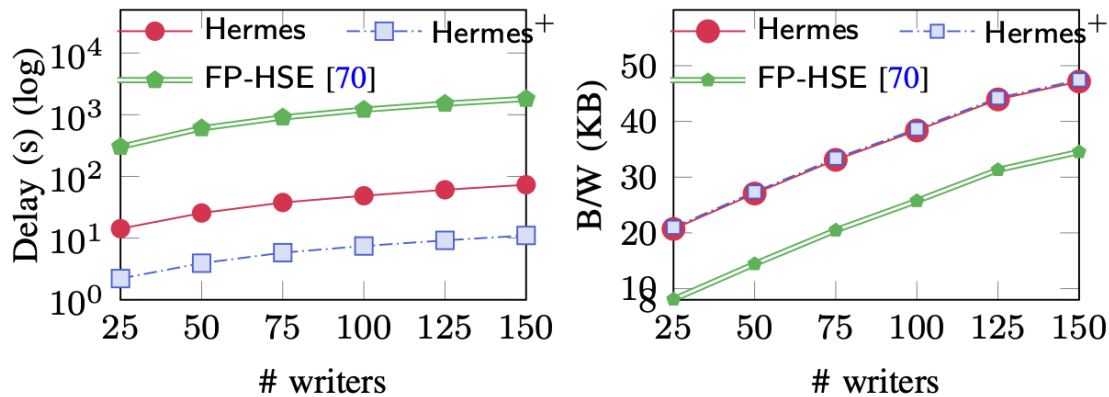
- Optimize search complexity to  $O(\sqrt{W})$ ,

and to  $O\left(\frac{\log^2(W)}{\log(\log(W))}\right)$  with recursion



## End-to-end keyword search delay

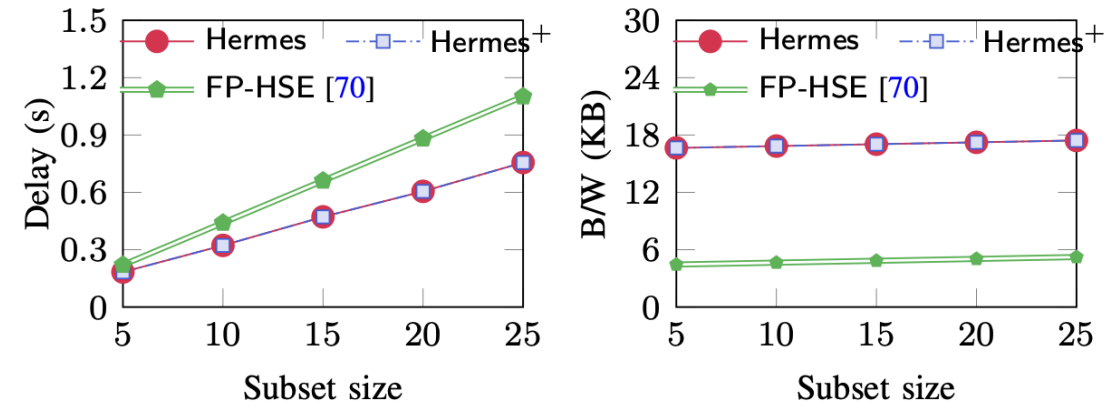
Hermes executes search up to **21.2×-163.8×** faster than FP-HSE [USENIX'22] while **preventing keyword-guessing attacks**



(a) E2E delay

(b) Reader's bandwidth

**Figure 7:** E2E keyword search delay and bandwidth (Enron).



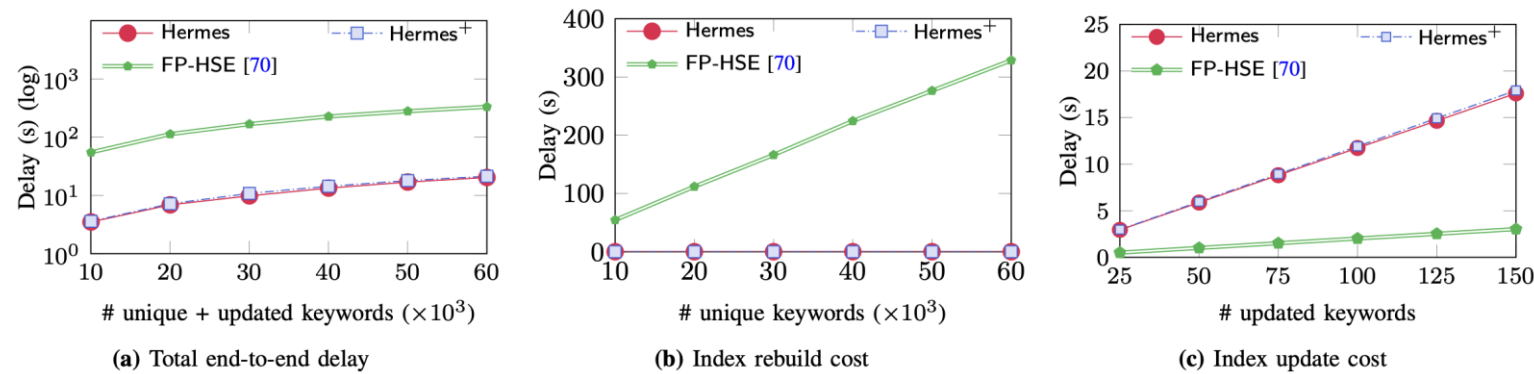
(a) E2E delay

(b) Reader's bandwidth

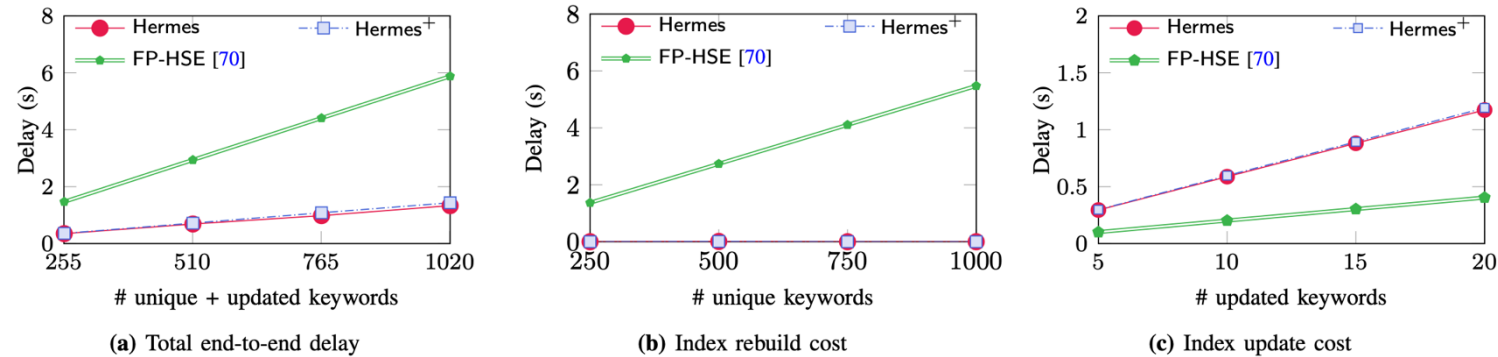
**Figure 8:** E2E keyword search delay and bandwidth (EHR).

## End-to-end keyword update delay

Hermes executes update **3.8×-17.1×** faster than FP-HSE

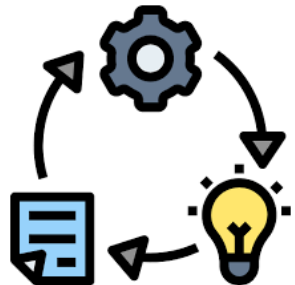


**Figure 10:** Keyword update delay (Enron). The total update cost with forward privacy (FP) (a) equals the cost to rebuild the index (b) plus the cost to update the index (c). Unlike FP-HSE, Hermes/Hermes<sup>+</sup> does not need a rebuild process to ensure FP of subsequent updates.



**Figure 11:** Keyword update delay (EHR).

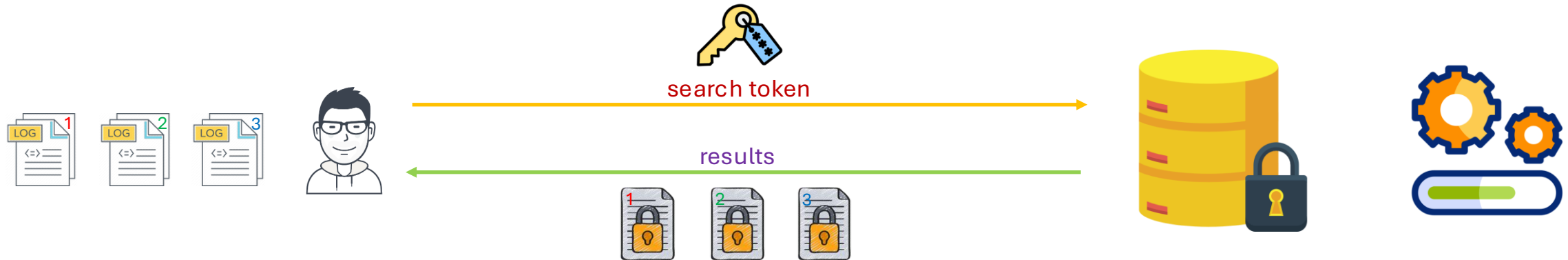
- Applying differential privacy to prevent leakage-abuse attacks more efficiently, without using MPC or costly cryptographic tools such as ORAM-based techniques
- Keyword search can complete in a single communication round-trip and output the actual identifiers of the matching documents
- Achieve small user-side storage



|      | CLRZ [INFOCOM'18]                                                                            | OSSE [NDSS'21]                                | <u>Our FROST</u>                                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pros | Fast with little overhead since DP is applied when database is outsourced                    | Output fresh obfuscation per query            | <ul style="list-style-type: none"> <li>Output fresh obfuscation per query</li> <li><b>Better performance:</b> up to four orders of magnitude faster and three orders of magnitude smaller bandwidth cost</li> </ul> |
| Cons | Produce identical obfuscated outputs for the same keyword, thereby revealing search patterns | High computational and communication overhead | Increased server storage cost                                                                                                                                                                                       |

- OSSE [NDSS'21] obfuscates search and result patterns by creating additional encrypted search tokens

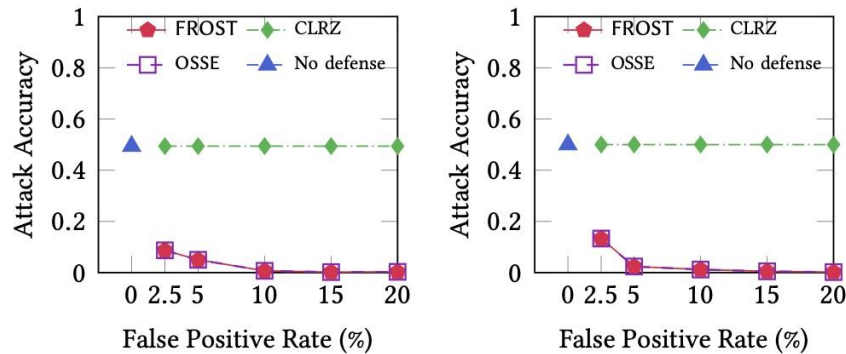
**Problem:** incur a lot of communication and computation overhead



- **Our idea:** obfuscation without additional processing for extra tokens
- **Techniques:**
  - Rerandomization PIR
  - State and transformation vectors



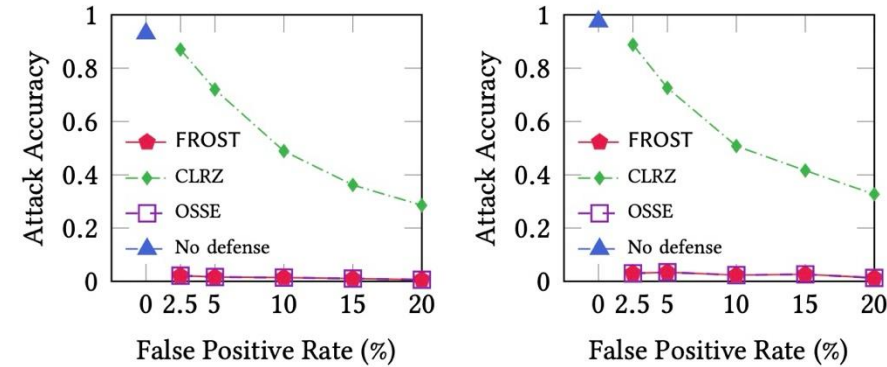
**Significantly reduce** the effectiveness of statistical leakage-abuse attacks, from 97.5%-49.4% down to **nearly random-guessing** (3.1%-0.2%)



(a) Enron

(b) Lucene

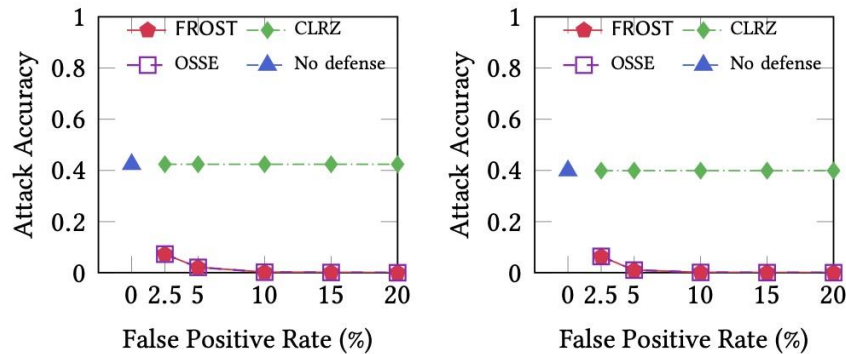
Figure 5: Accuracy of Freq attack ( $|W| = 1000$ ).



(a) Enron

(b) Lucene

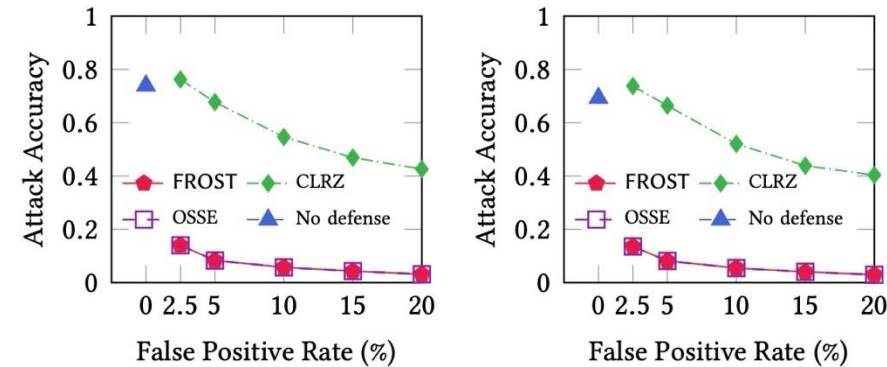
Figure 9: Accuracy of IHOP attack ( $|W| = 1000$ ).



(a)  $|W| = 3000$

(b)  $|W| = 5000$

Figure 6: Accuracy of Freq attack (Wiki dataset).



(a)  $|W| = 3000$

(b)  $|W| = 5000$

Figure 10: Accuracy of IHOP attack (Wiki dataset).

Execute keyword search **460×-10012× faster** with **1152×-3748× smaller bandwidth overhead** than the state-of-the-art

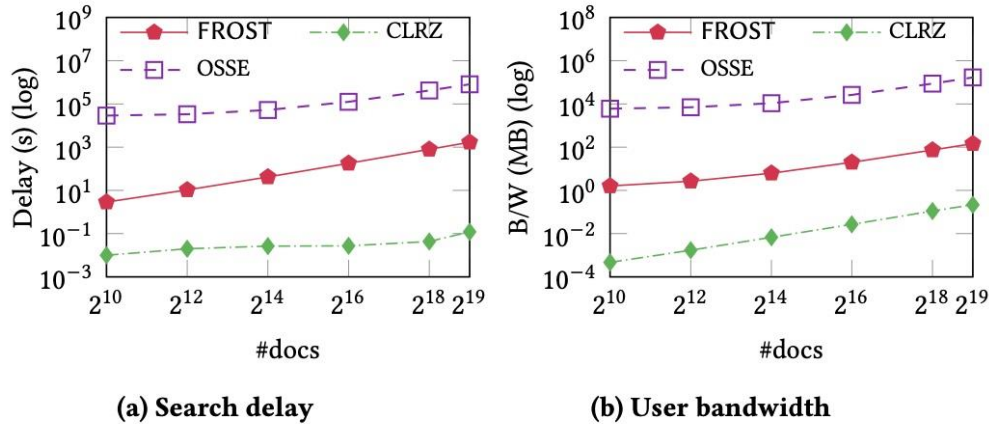


Figure 11: Total search delay and bandwidth (Enron, FPR = 0.1).

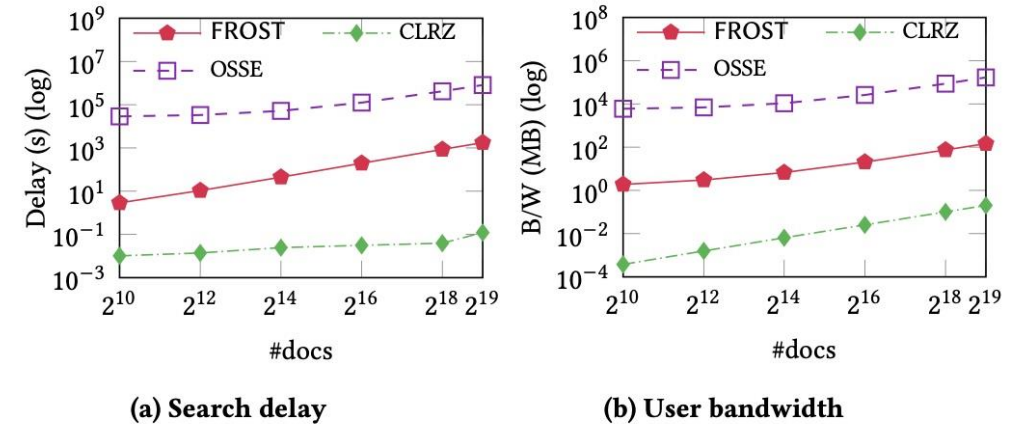


Figure 13: Total search delay and bandwidth (Wiki, FPR = 0.1).

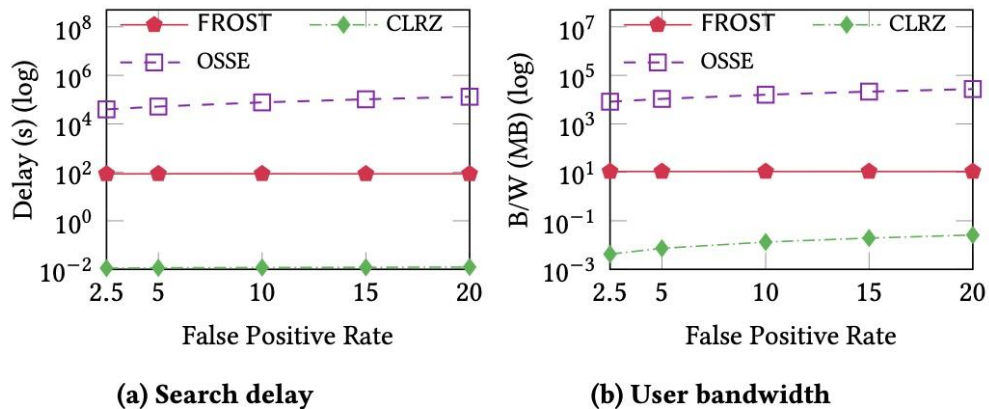


Figure 12: Total search delay and bandwidth (Enron, 32K documents).

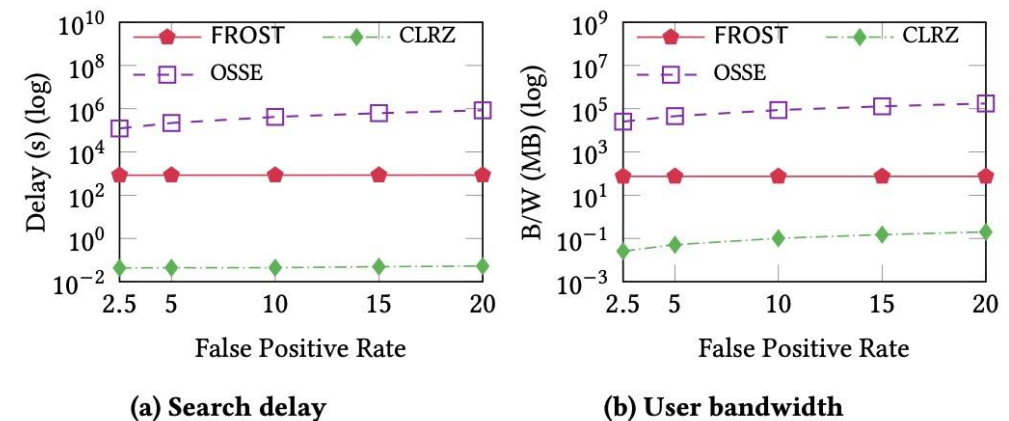


Figure 14: Total search delay and bandwidth (Wiki, 256K documents).

- In summary, this dissertation:



- Propose an **efficient solution** for **verifying** the **integrity** of outsourced data, addressing a long-standing and critical open issue
- Address user/data privacy and utilization dilemma: provide **efficient search functionality** while preventing information leakage



All are essential to build **practical encrypted data outsourcing systems** providing **high performance** and **security guarantees**



# THANK YOU FOR YOUR ATTENTION

## Q&A





**VIRGINIA TECH.®**

- [TPAMI'98] S. Umeyama, "An eigendecomposition approach to weighted graph matching problems," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 10, no. 5, pp. 695-703, Sept. 1988.
- [S&P'00] Dawn Xiaoding Song, D. Wagner, and A. Perrig. Practical techniques for searches on encrypted data. In *Proceeding 2000 IEEE Symposium on Security and Privacy*. S&P 2000, pages 44–55, 2000.
- [EUROCRYPT'01] Dan Boneh, and Matthew Franklin. Identity Based Encryption From the Weil Pairing. In *Proceedings of the 21st Annual International Cryptology Conference on Advances in Cryptology*, CRYPTO 2001.
- [EUROCRYPT'04] Dan Boneh, Giovanni Di Crescenzo, Rafail Ostrovsky, and Giuseppe Persiano. Public key encryption with keyword search. *Cryptology ePrint Archive*, Paper 2003/195, 2003. <https://eprint.iacr.org/2003/195>.
- [CCS'06] Reza Curtmola, Juan Garay, Seny Kamara, and Rafail Ostrovsky. Searchable symmetric encryption: Improved definitions and efficient constructions. In *Proceedings of the 13th ACM Conference on Computer and Communications Security*, CCS '06, page 79–88, New York, NY, USA, 2006. Association for Computing Machinery.
- [SDM'06] Jin Wook Byun, Hyun Suk Rhee, Hyun-A Park, and Dong Hoon Lee. Off-line keyword guessing attacks on recent keyword search schemes over encrypted data. In *Proceedings of the Third VLDB international conference on Secure Data Management (SDM'06)*.
- [CCS'07] Giuseppe Ateniese, Randal Burns, Reza Curtmola, Joseph Herring, Lea Kissner, Zachary Peterson, and Dawn Song. Provable data possession at untrusted stores. In *Proceedings of the 14th ACM conference on Computer and communications security*, pages 598–609, 2007.
- [CCS'07] Ari Juels and Burton S Kaliski Jr. Pors: Proofs of retrievability for large files. In *Proceedings of the 14th ACM conference on Computer and communications security*, pages 584–597, 2007.
- [CCSW'09] Kevin D Bowers, Ari Juels, and Alina Oprea. Proofs of retrievability: Theory and implementation. In *Proceedings of the 2009 ACM workshop on Cloud computing security*, pages 43–54, 2009.
- [TPAMI'09] M. Zaslavskiy, F. Bach and J. -P. Vert, "A Path Following Algorithm for the Graph Matching Problem," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 31, no. 12, pp. 2227-2242, Dec. 2009.
- [CRYPTO'11] Siavosh Benabbas, Rosario Gennaro, and Yevgeniy Vahlis. Verifiable delegation of computation over large datasets. In *Annual Cryptology Conference*, pages 111–131. Springer, 2011.

- [ACSAC'12] Emil Stefanov, Marten van Dijk, Ari Juels, and Alina Oprea. Iris: A scalable cloud file system with efficient integrity checks. In *ACSAC'12*, pages 229–238, 2012.
- [CCS'12] Seny Kamara, Charalampos Papamanthou, and Tom Roeder. Dynamic searchable symmetric encryption. In *Proceedings of the 2012 ACM conference on Computer and communications security (CCS '12)*.
- [NDSS'12] Mohammad Saiful Islam, Mehmet Kuzu, and Murat Kantarcioglu. “Access pattern disclosure on searchable encryption: ramification, attack and mitigation”. In *NDSS*, 2012.
- [CCS'12] Seny Kamara, Charalampos Papamanthou, and Tom Roeder. Dynamic searchable symmetric encryption. In *Proceedings of the 2012 ACM conference on Computer and communications security (CCS '12)*.
- [NDSS'12] Mohammad Saiful Islam, Mehmet Kuzu, and Murat Kantarcioglu. “Access pattern disclosure on searchable encryption: ramification, attack and mitigation”. In *NDSS*, 2012.
- [CCS'13] Elaine Shi, Emil Stefanov, and Charalampos Papamanthou. Practical dynamic proofs of retrievability. In *ACM CCS'13*, pages 325–336, 2013.
- [CRYPTO'13] Dan Boneh, Kevin Lewi, Hart William Montgomery, and Ananth Raghunathan. Key homomorphic prfs and their applications. In *Annual International Cryptology Conference*, 2013.
- [NDSS'14] David Cash, Joseph Jaeger, Stanislaw Jarecki, Charanjit S. Jutla, Hugo Krawczyk, Marcel-Catalin Rosu, and Michael Steiner. Dynamic searchable encryption in very-large databases: Data structures and implementation. *IACR Cryptol. ePrint Arch.*, 2014:853, 2014.
- [IS'14] Chang Liu, Liehuang Zhu, Mingzhong Wang, and Yuan Tan. Search pattern leakage in searchable encryption: Attacks and new construction. In *Information Sciences*, Volume 265, 2014, Pages 176-188.
- [TPDS'14] C. -K. Chu, S. S. M. Chow, W. -G. Tzeng, J. Zhou and R. H. Deng, "Key-Aggregate Cryptosystem for Scalable Data Sharing in Cloud Storage," in *IEEE Transactions on Parallel and Distributed Systems*, vol. 25, no. 2, pp. 468-477, Feb. 2014.
- [CCS'15] David Cash, Paul Grubbs, Jason Perry, and Thomas Ristenpart. Leakage-abuse attacks against searchable encryption. In *CCS*, 2015.
- [CCS'16] David Pouliot and Charles V Wright. The shadow nemesis: Inference attacks on efficiently deployable, efficiently searchable encryption. In *CCS*, 2016.

- [CCS'16] Raphael Bost.  $\Sigma\phi\phi\phi$ : Forward secure searchable encryption. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, CCS '16, page 1143–1154, New York, NY, USA, 2016. Association for Computing Machinery.
- [CCS'17] Raphaël Bost, Brice Minaud, and Olga Ohrimenko. Forward and Backward Private Searchable Encryption from Constrained Cryptographic Primitives. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS '17)*.
- [JoC'17] David Cash, Alptekin Kupcu, and Daniel Wichs. Dynamic proofs of retrievability via oblivious ram. *Journal of Cryptology*, 2017.
- [PET'19] Thang Hoang, Muslum Ozgur Ozmen, Yeongjin Jang, and Attila A Yavuz. Hardware-supported oram in effect: Practical oblivious search and update on very large dataset. *Proceedings on Privacy Enhancing Technologies*, 2019(1), 2019.
- [OSDI'20] Emma Dauterman, Eric Feng, Ellen Luo, Raluca Ada Popa, and Ion Stoica. Dory: An encrypted search system with distrust. In *Proceedings of the 14th USENIX Conference on Operating Systems Design and Implementation*, OSDI'20, USA, 2020. USENIX Association.
- [NDSS'20] Laura Blackstone, Seny Kamara, and Tarik Moataz. Revisiting leakage abuse attacks. In *NDSS*, 2020.
- [CCS'21] Jianting Ning, Xinyi Huang, Geong Sen Poh, Jiaming Yuan, Yingjiu Li, Jian Weng, and Robert H Deng. Leap: Leakage-abuse attack on efficiently deployable, efficiently searchable encryption with partially known dataset. In *CCS*, 2021.
- [USENIX'21] Marc Damie, Florian Hahn, and Andreas Peter. A highly accurate query-recovery attack against searchable encryption using non-indexed documents. In *USENIX Security*, 2021.
- [USENIX'21] Simon Oya and Florian Kerschbaum. Hiding the access pattern is not enough: Exploiting search pattern leakage in searchable encryption. In *USENIX Security*, 2021.
- [USENIX'21] Gaspard Anthoine, Jean-Guillaume Dumas, Mélanie de Jonghe, Aude Maignan, and Clément Pernet, Michael Hanling, and Daniel S Roche. Dynamic proofs of retrievability with low server storage. In *30th USENIX Security Symposium*, pages 537–554, 2021.
- [USENIX'22] Simon Oya and Florian Kerschbaum. IHOP: Improved statistical query recovery against searchable symmetric encryption through quadratic optimization. In *USENIX Security*, 2022.
- [USENIX'22] Jiafan Wang and Sherman S. M. Chow. Omnes pro uno: Practical Multi-Writer encrypted database. In *31st USENIX Security Symposium (USENIX Security 22)*, pages 2371–2388, Boston, MA, August 2022. USENIX Association.

- [PKC'22] Ryo Nishimaki. 2022. The Direction of Updatable Encryption Does Matter. In Public-Key Cryptography – PKC 2022: 25th IACR International Conference on Practice and Theory of Public-Key Cryptography.
- [TCC'22] Y. Miao, Q. Tong, R. H. Deng, K. -K. R. Choo, X. Liu and H. Li, "Verifiable Searchable Encryption Framework Against Insider Keyword-Guessing Attack in Cloud Storage," in *IEEE Transactions on Cloud Computing*, vol. 10, no. 2, pp. 835-848, 1 April-June 2022.
- [CCS'23] Lei Xu, Leqian Zheng, Chengzhi Xu, Xingliang Yuan, and Cong Wang. 2023. Leakage-Abuse Attacks Against Forward and Backward Private Searchable Symmetric Encryption. In CCS '23.
- [TCC'23] Daniel Slamanig and Christoph Striecks. Revisiting Updatable Encryption: Controlled Forward Security, Constructions and a Puncturable Perspective. In Theory of Cryptography: 21st International Conference, TCC 2023.
- [PETS'23] Tung Le, and Thang Hoang. "MAPLE: A Metadata-Hiding Policy-Controllable Encrypted Search Platform with Minimal Trust." In 2023 Privacy Enhancing Technologies Symposium (*PETS 2023*), Lausanne, Switzerland, July 2023.
- [CODASPY'24] Jacob Haltiwanger and Thang Hoang. Exploiting Update Leakage in Searchable Symmetric Encryption. In Proceedings of the Fourteenth ACM Conference on Data and Application Security and Privacy (CODASPY '24).
- [USENIX'24] Tung Le, Rouzbeh Behnia, Jorge Guajardo, and Thang Hoang. "MUSES: Efficient Multi-User Searchable Encrypted Database." In USENIX Security Symposium (*USENIX Security 2024*), Philadelphia, PA, August 2024.
- [USENIX'24] Hao Nie and Wei Wang and Peng Xu and Xianglong Zhang and Laurence T. Yang and Kaitai Liang. "Query Recovery from Easy to Hard: Jigsaw Attack against SSE." In USENIX Security Symposium (*USENIX Security 2024*), Philadelphia, PA, August 2024.
- [S&P'25] Tung Le, and Thang Hoang. "Hermes: Efficient and Secure Multi-Writer Encrypted Database," *2025 IEEE Symposium on Security and Privacy (SP)*, San Francisco, CA, USA, 2025.